SHIV NADAR UNIVERSITY

Program: **B.Tech** Discipline: **Computer Science and Engineering** Exam: **Quiz 1**
Year: **2018-2019** Course Code: **CSD207** Course Title: **Object Oriented Programming in Java**
Date: **September 6, 2018** Duration: **1 hour** Max. Marks: **30**
Name: SNU ID: Signature:

Instructions: 1. This quiz has weightage of 5% in your overall 100 points.
2. Circle the correct answer.
3. There are 30 questions in this quiz. A question may have more than one correct answer.

1. Which of the following is a valid identifier?

a. $343
b. class
c. 9X
d. 8+9
e. radius

Key : a, e

class is a keyword, which cannot be used as an identifier. Identifiers cannot start with a number.

2. The expression (int)(76.0252175 * 100) / 100 evaluates to _____.

a. 76.02
b. 76
c. 76.0252175
d. 76.03

Key : b

In order to obtain 76.02, you have divide 100.0.

3. Which of the following are correct names for variables according to Java naming conventions?

a. radius
b. Radius
c. RADIUS
d. findArea
e. FindArea

Key : a, d

A single-word variable is in lowercase. In a multiple-word variable, the words are concatenated with the first word in lowercase and the first letter of each subsequent word in uppercase.

4. Which of the following assignment statements is illegal?

a. float f = -34;
b. int t = 23;
c. short s = 10;
d. int t = (int)false;
e. int t = 4.5;

Key : d, e

5. What is the output from System.out.println((int)Math.random() * 4)?

a. 0
b. 1
c. 2
d. 3
e. 4

Key : a

Casting is performed before the * operator in (int)Math.random() * 4. So, it returns 0.

6. Suppose x=10 and y=10. What is x after evaluating the expression (y >= 10) || (x++ > 10).

a. 9
b. 10
c. 11

Key : b

For the || operator, the right operand is not evaluated, if the left operand is evaluated as true

7. What is 1 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1 == 0.5?

a. true
b. false
c. There is no guarantee that 1 - 0.1 - 0.1 - 0.1 - 0.1 - 0.1 == 0.5 is true.

Key : c

This epression involves floating-point number. Floating-point numbers are approximated. The correct answer is C.

8. What is the output of the following code?

```
boolean even = false;
System.out.println((even ? "true" : "false"));
```

a. true
b. false
c. nothing
d. true false

Key : b

Since even is false, the conditional expression yields false. The correct answer is B.

9. Which of the following statements are correct?

a. char[][] charArray = {'a', 'b'};
b. char[2][2] charArray = {{'a', 'b'}, {'c', 'd'}};
c. char[2][] charArray = {{'a', 'b'}, {'c', 'd'}};
d. char[][] charArray = {{'a', 'b'}, {'c', 'd'}};

Key : d

10. Does the method call in the following method cause compile errors?

```
public static void main(String[] args) {
  Math.pow(2, 4);
}
```

a. Yes
b. No

Key : b

A value-returning method can also be invoked as a statement in Java. In this case, the caller simply ignores the return value. This is rare, but permissible if the caller is not interested in the return value.

11. What is Math.rint(3.5)?

a. 3.0
b. 3
c. 4
d. 4.0
e. 5.0

Key : d

rint returns the nearest even integer as a double since 3.5 is equally close to 3.0 and 4.0.

12. Consider the following incomplete code:

```
public class Test {
  public static void main(String[] args) {
    System.out.println(f(5));
  }
  public static int f(int number) {
    // Missing body
  }
}
```

The missing method body should be _____.
a. return "number";
b. System.out.println(number);
c. System.out.println("number");
d. return number;

Key : d

13. Which of the following statements are correct?

a. char[][][] charArray = new char[2][2][];

b. char[2][2][] charArray = {'a', 'b'};

c. char[][][] charArray = {{'a', 'b'}, {'c', 'd'}, {'e', 'f'}};

d. char[][][] charArray = {{{'a', 'b'}, {'c', 'd'}, {'e', 'f'}}};

Key : a, d

14. Analyze the following code.

```
int[] list = new int[5];
list = new int[6];
```

a. The code has compile errors because the variable list cannot be changed once it is assigned.

b. The code has runtime errors because the variable list cannot be changed once it is assigned.

c. The code can compile and run fine. The second line assigns a new array to list.

d. The code has compile errors because you cannot assign a different size array to list.

Key : c

15. Which of the following is the best for generating random integer between 0 or 1?

a. (int)Math.random()

b. (int)Math.random() + 1

c. (int)(Math.random() + 0.5)

d. (int)(Math.random() + 0.2)

e. (int)(Math.random() + 0.8)

Key : c

16. Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    boolean[][] x = new boolean[3][];
    x[0] = new boolean[1]; x[1] = new boolean[2];
    x[2] = new boolean[3];
    System.out.println("x[2][2] is " + x[2][2]);
  }
}
```

a. The program has a compile error because new boolean[3][] is wrong.
b. The program has a runtime error because x[2][2] is null.
c. The program runs and displays x[2][2] is null.
d. The program runs and displays x[2][2] is true.
e. The program runs and displays x[2][2] is false.

Key : e

x is a ragged array. (See the section on Ragged Array) x[2] has three elements with default value false.

17. Assume int[][] x = {{1, 2}, {3, 4}, {5, 6}}, what are x.length are x[0].length?

a. 2 and 1
b. 2 and 2
c. 3 and 2
d. 2 and 3
e. 3 and 3

Key : c

18. Analyze the following code:

```
public class Test {
  public static void main(String[] args) {
    System.out.println(xMethod(5, 500L));
  }
  public static int xMethod(int n, long l) {
    System.out.println("int, long");
    return n;
  }
  public static long xMethod(long n, long l) {
    System.out.println("long, long");
    return n;
  }
}
```

a. The program displays int, long followed by 5.
b. The program displays long, long followed by 5.
c. The program runs fine but displays things other than 5.
d. The program does not compile because the compiler cannot distinguish which xmethod to invoke.

Key : a

19. Which of the following is the correct statement to return JAVA?

a. toUpperCase("Java")
b. "Java".toUpperCase("Java")
c. "Java".toUpperCase()
d. String.toUpperCase("Java")

Key : c

The correct method is toUpperCase(). So C is correct.

20. Suppose s is a string with the value "java". What will be assigned to x if you execute the following code?

char x = s.charAt(4);

a. 'a'
b. 'v'
c. Nothing will be assigned to x, because the execution causes the runtime error StringIndexOutofBoundsException.

Key : c

The string index starts from 0 and the last index is s.length() - 1. s.charAt(4) is out of bounds.

21. What is the return value of "SELECT".substring(4, 4)?

a. an empty string
b. C
c. T
d. E

Key : a

If beginIndex is endIndex, substring(beginIndex, endIndex) returns an empty string with length 0. It would be a runtime error, if beginIndex > endIndex.

22. What is output of the following code:

```
public class Test {
  public static void main(String[] args) {
    int list[] = {1, 2, 3, 4, 5, 6};
    for (int i = 1; i < list.length; i++)
      list[i] = list[i - 1];
    for (int i = 0; i < list.length; i++)
      System.out.print(list[i] + " ");
  }
}
```

a. 1 2 3 4 5 6
b. 2 3 4 5 6 6
c. 2 3 4 5 6 1
d. 1 1 1 1 1 1

Key : d

23. What is the output of the following program?

```
public class Test {
  public static void main(String[] args) {
    int[][] values = {{3, 4, 5, 1}, {33, 6, 1, 2}};
    int v = values[0][0];
    for (int row = 0; row < values.length; row++)
      for (int column = 0; column < values[row].length;
column++)
        if (v < values[row][column])
          v = values[row][column];
    System.out.print(v);
  }
}
```

a. 1
b. 3
c. 5
d. 6
e. 33

Key : e

24. What is the output of the following program?

```
public class Test {
  public static void main(String[] args) {
    int[][] values = {{3, 4, 5, 1 }, {33, 6, 1, 2}};
    for (int row = 0; row < values.length; row++) {
      java.util.Arrays.sort(values[row]);
      for (int column = 0; column < values[row].length;
column++)
        System.out.print(values[row][column] + " ");
      System.out.println();
    }
  }
}
```

a. The program prints two rows 3 4 5 1 followed by 33 6 1 2
b. The program prints on row 3 4 5 1 33 6 1 2
c. The program prints two rows 3 4 5 1 followed by 2 1 6 33
d. The program prints two rows 1 3 4 5 followed by 1 2 6 33
e. The program prints one row 1 3 4 5 1 2 6 33

Key : d

25. Show the output of the following code:

```
public class Test {
  public static void main(String[] args) {
    int[] x = {1, 2, 3, 4, 5};
    increase(x);
    int[] y = {1, 2, 3, 4, 5};
    increase(y[0]);
    System.out.println(x[0] + " " + y[0]);
  }
  public static void increase(int[] x) {
    for (int i = 0; i < x.length; i++)
      x[i]++;
  }
  public static void increase(int y) {
    y++;
  }
}
```

a. 0 0
b. 1 1
c. 2 2
d. 2 1
e. 1 2

Key : d

Invoking increase(x) passes the reference of the array to the method. Invoking increase(y[0]) passes the value 1 to the method. The value y[0] outside the method is not changed.

26. Analyze the following code:

```
public class Test1 {
```

```
  public static void main(String[] args) {
    xMethod(new double[]{3, 3});
    xMethod(new double[5]);
    xMethod(new double[3]{1, 2, 3});
  }
  public static void xMethod(double[] a) {
    System.out.println(a.length);
  }
}
```

a. The program has a compile error because xMethod(new double[]{3, 3}) is incorrect.
b. The program has a compile error because xMethod(new double[5]) is incorrect.
c. The program has a compile error because xMethod(new double[3]{1, 2, 3}) is incorrect.
d. The program has a runtime error because a is null.

Key : c

new double[3]{1, 2, 3} should be replaced by new double[]{1, 2, 3}) (anonymous array).

27. Analyze the following code.

```
public class Test {
  public static void main(String[] args)  {
    System.out.println(max(1, 2));
  }
  public static double max(int num1, double num2) {
    System.out.println("max(int, double) is invoked");
    if (num1 > num2)
      return num1;
    else
      return num2;
  }
  public static double max(double num1, int num2) {
    System.out.println("max(double, int) is invoked");
    if (num1 > num2)
      return num1;
    else
      return num2;
  }
}
```

a. The program cannot compile because you cannot have the print statement in a non-void method.
b. The program cannot compile because the compiler cannot determine which max method should be invoked.
c. The program runs and prints 2 followed by "max(int, double)" is invoked.
d. The program runs and prints 2 followed by "max(double, int)" is invoked.
e. The program runs and prints "max(int, double) is invoked" followed by 2.

Key : b

This is known as ambiguous method invocation.

28. Analyze the following code.

```
public class Test {
  public static void main(String[] args)  {
    System.out.println(m(2));
  }
 public static int m(int num) {
    return num;
  }
  public static void m(int num) {
    System.out.println(num);
  }
}
```

a. The program has a compile error because the two methods m have the same signature.
b. The program has a compile error because the second m method is defined, but not invoked in the main method.
c. The program runs and prints 2 once.
d. The program runs and prints 2 twice.

Key : a

You cannot override the methods based on the type returned.

29. Analyze the following code:

```
import java.util.Scanner;
public class Test {
  public static void main(String[] args) {
    int sum = 0;
    for (int i = 0; i < 100000; i++) {
      Scanner input = new Scanner(System.in);
      sum += input.nextInt();
    }
  }
}
```

a. The program does not compile because the Scanner input = new Scanner(System.in); statement is inside the loop.
b. The program compiles, but does not run because the Scanner input = new Scanner(System.in); statement is inside the loop.
c. The program compiles and runs, but it is not efficient and unnecessary to execute the Scanner input = new

Scanner(System.in); statement inside the loop. You should move the statement before the loop.
d. The program compiles, but does not run because there is not prompting message for entering the input.

Key : c

To receive input from the keyboard, you need to create an input object from the Scanner class. You should create this object only once in the program. Placing the statement Scanner input = new Scanner(System.in) in the loop causes it to be created multiple times, which is a bad practice and could lead to potential errors. So, the correct answer is C.

30. Analyze the following code:

Code 1:

```
int number = 45;
boolean even;
if (number % 2 == 0)
  even = true;
else
  even = false;
```

Code 2:

```
int number = 45;
boolean even = (number % 2 == 0);
```

a. Code 1 has compile errors.
b. Code 2 has compile errors.
c. Both Code 1 and Code 2 have compile errors.
d. Both Code 1 and Code 2 are correct, but Code 2 is better.

Key : d

Both Code 1 and Code 2 are correct. Clearly Code 2 is shorter and better.